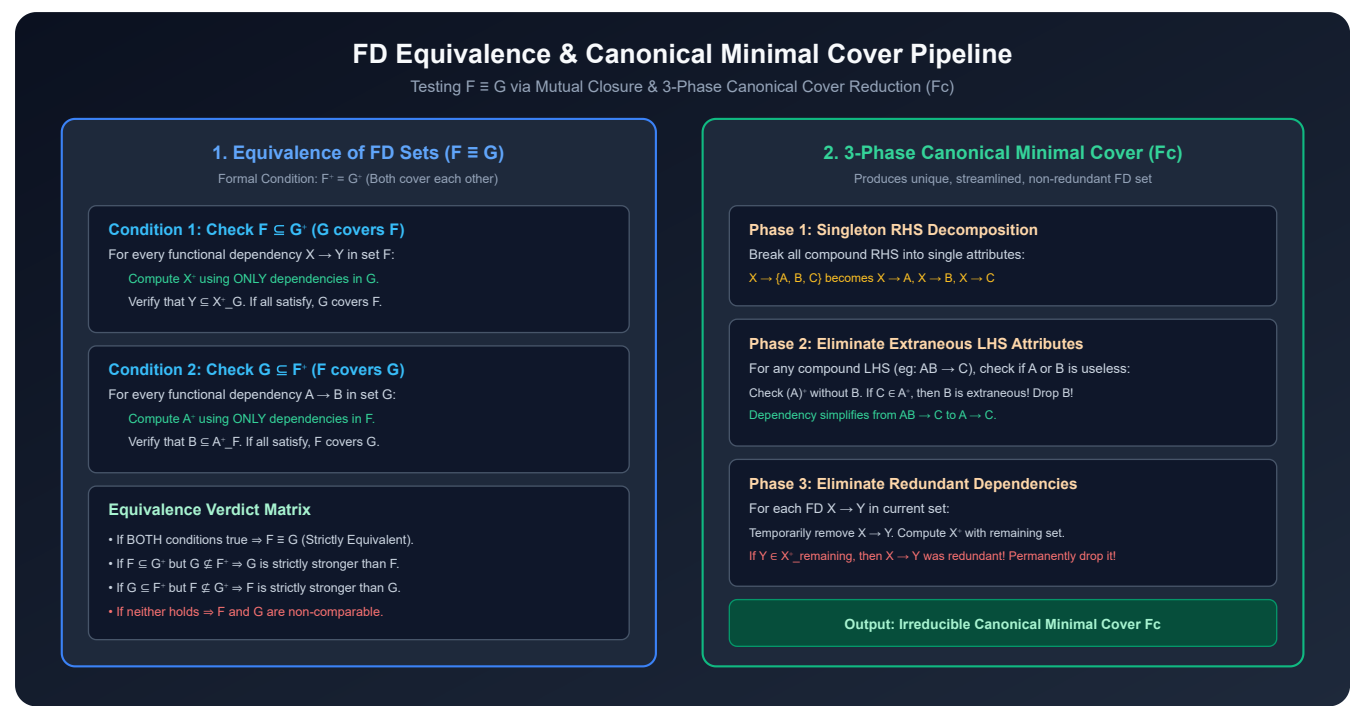


Module 04: Equivalence & Canonical Minimal Cover



$\rightarrow B \in G$:

- Compute the attribute closure of A using **ONLY the dependencies of F** (A^+_F).
- If $B \subseteq A^+_F$ holds for every FD in G , then F covers G .

3. Conclusion:

- If BOTH Test 1 and Test 2 pass $\implies \mathbf{F \equiv G}$.

2. Canonical Cover / Minimal Cover (F_c)

Database schema ko optimize karne ke liye hume aisa equivalent FD set chahiye jisme koi bhi extra (redundant) attribute ya rule na ho. Is minimal non-redundant set ko **Canonical Cover** (F_c) ya **Minimal Cover** kehte hain.

2.1 Formal Properties of a Canonical Cover

Ek set F_c canonical cover tabhi hota hai jab:

- 1. Singleton RHS:** Har FD ke right side par strictly **sirf ek attribute** ho (X

$\rightarrow A$).

- 2. No Extraneous LHS Attributes:** Left-hand side par koi faltu attribute na ho (e.g., agar A

$\rightarrow B$ se kaam chal raha hai toh AC

$\rightarrow B$ me C extraneous hai).

- 3. No Redundant Dependencies:** Koi aisi dependency na ho jisko hata dene par bhi closure me koi farq na pade.

3. The 3-Phase Canonical Cover Algorithm

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Step 1: Right-Hand Side Decomposition

RHS ke sabhi multi-attributes ko singleton me tod do.

Eg: $X \rightarrow ABC$ becomes $\{X \rightarrow A, X \rightarrow B, X \rightarrow C\}$

Step 2: Extraneous Attribute Removal (from LHS)

Har composite LHS dependency (eg: $AB \rightarrow C$) ke liye check karo:

- Kya A ko hataya ja sakta hai? Calculate $(B)^+$. If $C \in (B)^+$, then A is extraneous!
- Kya B ko hataya ja sakta hai? Calculate $(A)^+$. If $C \in (A)^+$, then B is extraneous!
- Extraneous attribute ko permanently delete karo.

Step 3: Redundant Dependency Removal

Har remaining dependency $X \rightarrow Y$ ke liye:

- Us dependency ko temporarily set se bahar nikalo: $F' = F - \{X \rightarrow Y\}$.
- Remaining set F' ko use karke X^+ calculate karo.
- Agar $Y \in X^+_{F'}$, iska matlab $X \rightarrow Y$ baaki bachi FDs se already derive ho raha hai!
- Yeh redundant hai! Isko permanently delete karo!

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4. Architectural Diagram

5. Gateway Classes Solved AKTU Examination Numericals

Numerical 1 (Full Canonical Cover Calculation)

Question: Given relation $R(A, B, C)$ and FD set:

$$F = \{ A$$

$$\rightarrow BC, \quad B$$

$$\rightarrow C, \quad A$$

$$\rightarrow B, \quad AB$$

$$\rightarrow C \}$$

Find the Canonical Cover (F_c).

Step 1: Singleton RHS Decomposition

$$A$$

$$\rightarrow BC \text{ splits into } A$$

$$\rightarrow B \text{ and } A$$

$$\rightarrow C.$$

Current set:

$$F_1 = \{ A$$

$$\rightarrow B, \quad A$$

$$\rightarrow C, \quad B$$

$$\rightarrow C, \quad A$$

$$\rightarrow B, \quad AB$$

$\rightarrow C \cup \{A\}$

Duplicate A

$\rightarrow B$ removed:

$F_1 = \{A\}$

$\rightarrow B, A$

$\rightarrow C, B$

$\rightarrow C, AB$

$\rightarrow C \cup \{A\}$

Step 2: Check Extraneous Attribute in AB

$\rightarrow C$

Check if B can be dropped from LHS:

- Compute closure of A without using AB

$\rightarrow C$:

- In $\{A\}$

$\rightarrow B, A$

$\rightarrow C, B$

$\rightarrow C \cup \{A\}, A^+ = \{A, B, C\}$.

- Since $C \in A^+$, A alone can derive C ! Therefore, **B is extraneous** in AB

$\rightarrow C$!

- Dependency AB

$\rightarrow C$ simplifies to A

$\rightarrow C$.

Since A

$\rightarrow C$ already exists in our set, duplicate is removed.

Set after Step 2:

$F_2 = \{A\}$

$\rightarrow B, A$

$\rightarrow C, B$

$\rightarrow C \cup \{A\}$

Step 3: Check Redundant Dependencies

1. Test A

$\Rightarrow B$:

- Remove it. Remaining: $\{A$

$\Rightarrow C, B$

$\Rightarrow C\}$.

- $A^+ = \{A, C\}$. Since B

$\in A^+, A$

$\Rightarrow B$ is **NOT redundant**. Keep it.

2. Test A

$\Rightarrow C$:

- Remove it. Remaining: $\{A$

$\Rightarrow B, B$

$\Rightarrow C\}$.

- $A^+ = \{A, B, C\}$ (using A

$\Rightarrow B$ and B

$\Rightarrow C$).

- Since $C \in A^+, A$

$\Rightarrow C$ is derived via transitivity!

- Therefore, A

$\Rightarrow C$ is REDUNDANT**! Remove it!

3. Test B

$\Rightarrow C$:

- Remove it. Remaining: $\{A$

$\Rightarrow B\}$.

- $B^+ = \{B\}$. Since C

$\in B^+, B$

$\Rightarrow C$ is **NOT redundant**. Keep it.

Final Answer:

$\mathbf{F_c} = \{ A$

$\Rightarrow B, \quad B$

$\rightarrow C \setminus \{x\}$